

Isothiocyanate concentrations and interconversion of sulforaphane to erucin in human subjects after consumption of commercial frozen broccoli compa...



SCOPE: Sulforaphane (a potent anticarcinogenic isothiocyanate derived from glucoraphanin) is widely considered responsible for the protective effects of broccoli consumption. Broccoli is typically purchased fresh or frozen and cooked before consumption. We compared the bioavailability and metabolism of sulforaphane from portions of lightly cooked fresh or frozen broccoli, and investigated the bioconversion of sulforaphane to erucin. METHODS AND RESULTS: Eighteen healthy volunteers consumed broccoli soups produced from fresh or frozen broccoli florets that had been lightly cooked and sulforaphane thio-conjugates quantified in plasma and urine. Sulforaphane bioavailability was about tenfold higher for the soups made from fresh compared to frozen broccoli, and the reduction was shown to be due to destruction of myrosinase activity by the commercial blanching-freezing process. Sulforaphane appeared in plasma and urine in its free form and as several thio-conjugates forms. Erucin N-acetyl-cysteine conjugate was a significant urinary metabolite, and it was shown that human gut microflora can produce sulforaphane, erucin, and their nitriles from glucoraphanin. CONCLUSION: The short period of blanching used to produce commercial frozen broccoli destroys myrosinase and substantially reduces sulforaphane bioavailability. Sulforaphane was converted to erucin and excreted in urine, and it was shown that human colonic flora were capable of this conversion. © 2012 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.